

CASE 01

Scaling drone delivery through navigation reliability

A product strategy for safer exception handling and route growth

Senior Product Manager · navigation and flight operations

Anonymised case study · 2023–2024 · photo from public news coverage



GOAL

Expand fixed delivery routes by 50% without weakening service quality

ORGANISATION

A 300-person drone delivery unit with a 25-person product team

MY ROLE

Diagnosed the constraint, defined three workstreams and led release across eight teams

OUTCOME

Takeover became reversible, rolled out in stages to 100 routes, then launched fleet-wide

Route growth made operator capacity the constraint

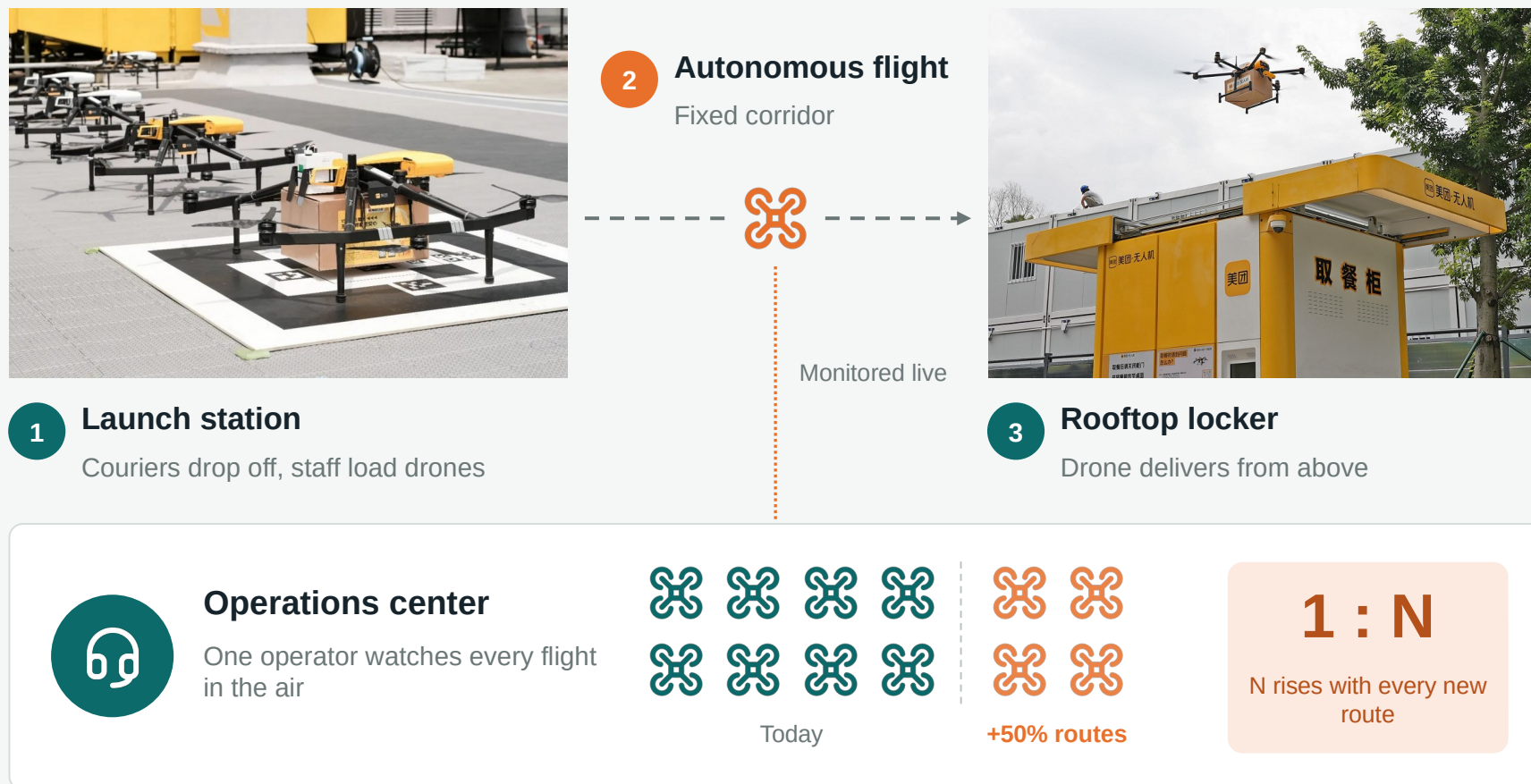
BUSINESS GOAL

+50%

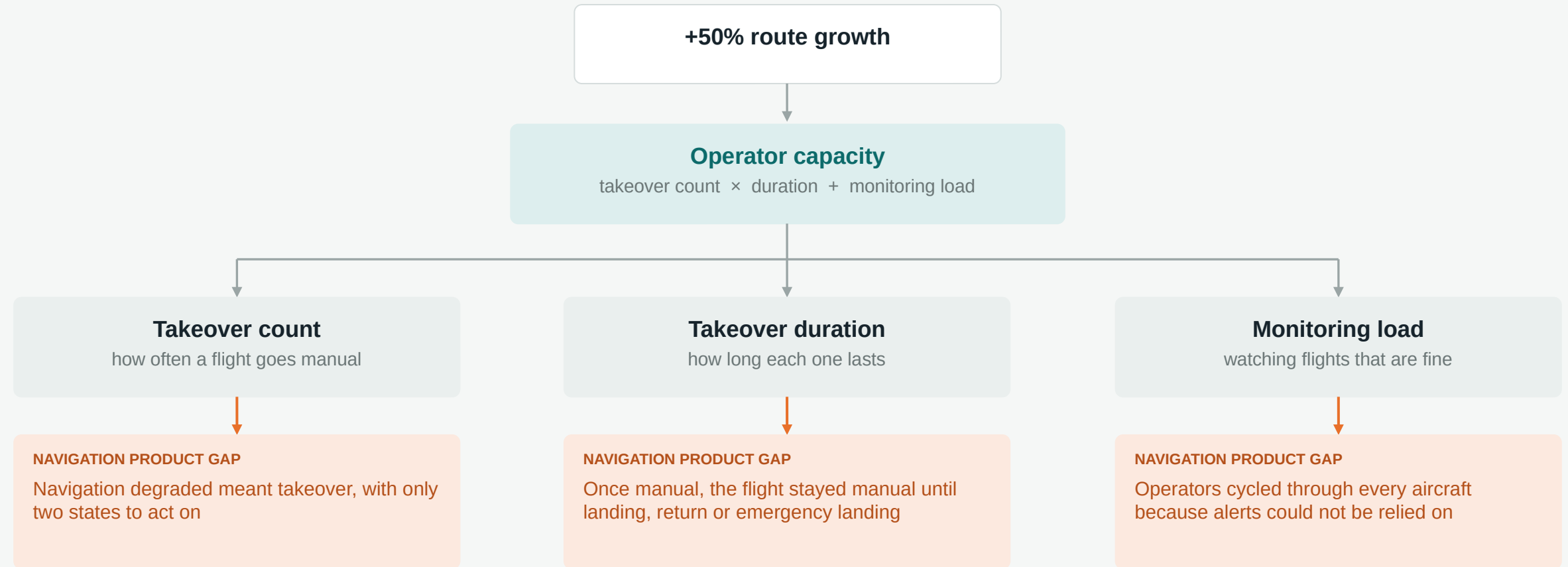
planned growth in fixed delivery routes

My brief

Leadership set the KPI. I found the constraint and defined the product work.



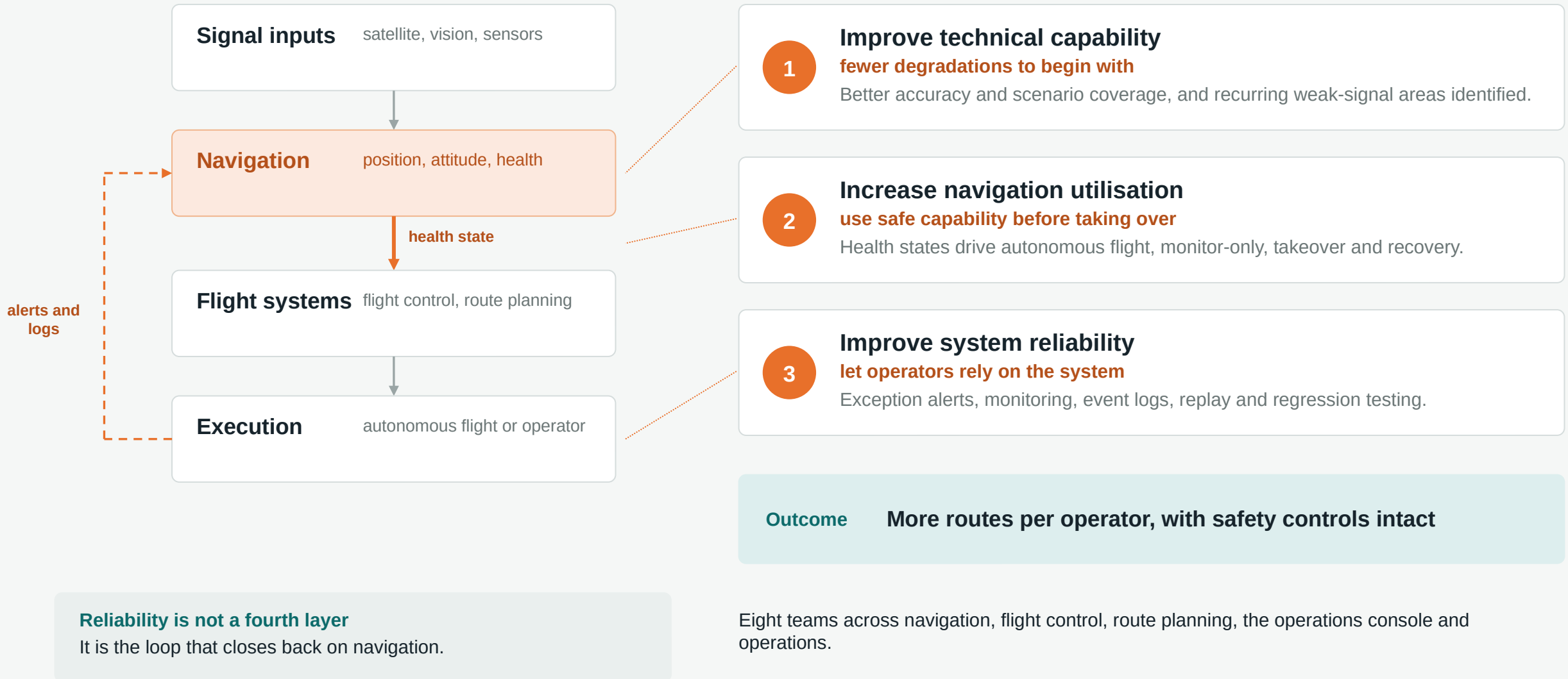
Route growth was limited by operator capacity



Evidence used positioning logs · flight incident reports · operator feedback · historical route data

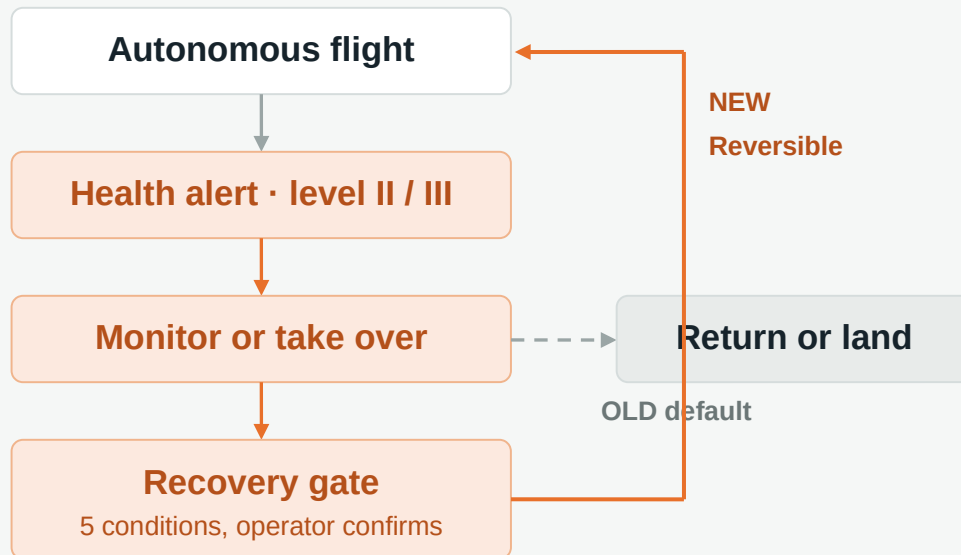
All three terms traced back to the same place: navigation reported a binary signal, so every degradation became a person.

Three workstreams, placed where navigation meets operations



Manual takeover became a reversible state

ONE STATE FLOW, TWO POLICIES



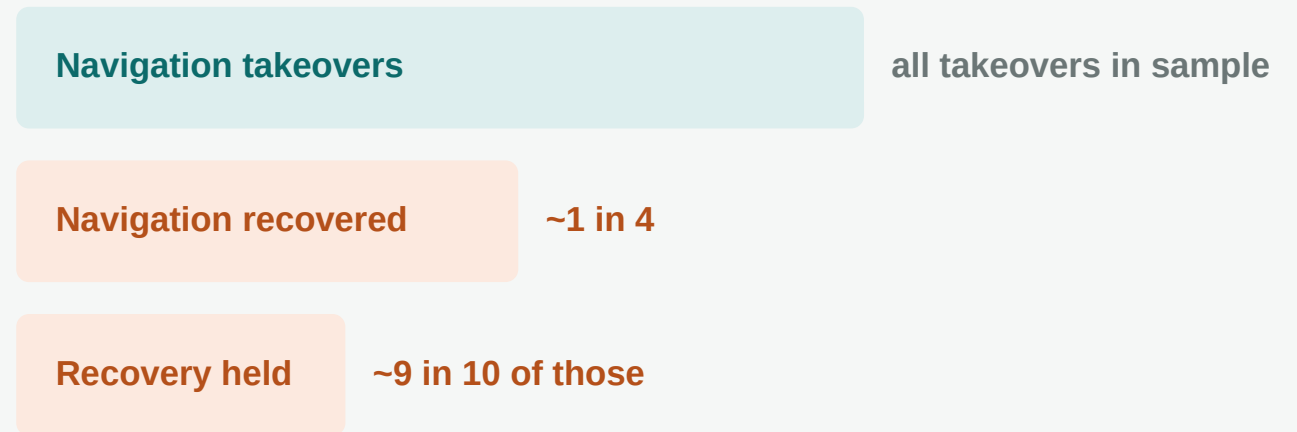
Why not automatic

Recovery is never automatic. Multi-source fix, accuracy, stable window, fresh route and an operator click are all required.

Proportions are approximate, from takeover log analysis used for the benefit case.

EVIDENCE BEHIND THE DECISION

Takeover log analysis

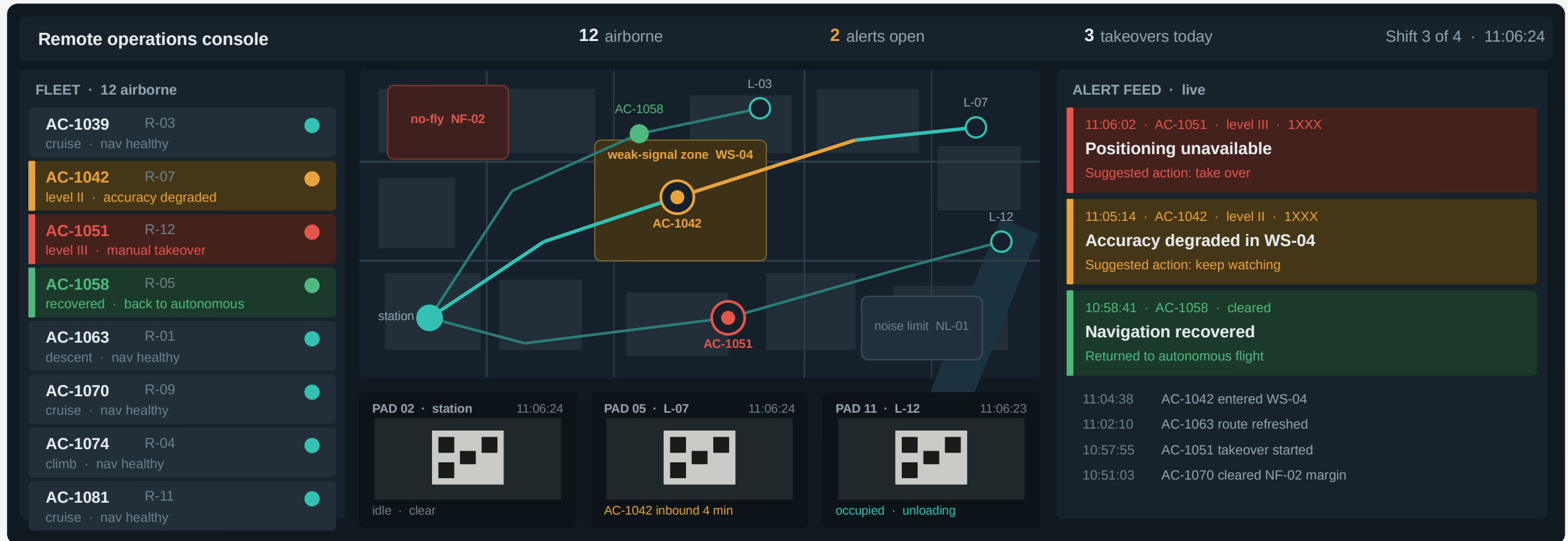


Roughly a quarter of takeovers involved navigation that had already recovered, and most of those recoveries held long enough to hand control back safely.

The trade-off

Engineering weighted safety and effort. Operations needed lower operator load. The funnel moved the debate from whether to build it to when and how to release it.

Navigation states became operator actions on screen

**LEVEL II****Stay autonomous**

Alert raised, protected envelope widened

FORCED MONITOR**Watch, do not fly**

Window cannot be closed until cleared

LEVEL III**Manual takeover**

Hold position or fly to a point on the map

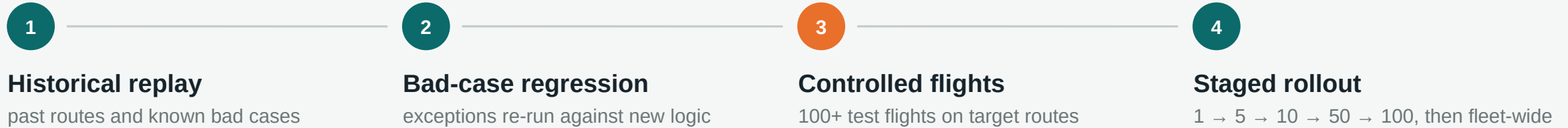
RECOVERED**Hand control back**

One click returns autonomy

What I defined: alert code to recommended action mapping · navigation entries in the existing three-level alert scheme · forced-monitor mode · return-to-autonomy control

Redrawn wireframe. Aircraft IDs, routes, zones, codes and times are illustrative; no internal screens, fields or log samples are shown.

Testing changed the design before routes were exposed



GATE METRICS AT EVERY STEP

Order success rate

Order delay rate

Blocking issues: severity and count

Three months from kick-off to 100 routes

A one-week delay was taken deliberately, before any route was exposed, rather than after.

WHAT CONTROLLED FLIGHTS EXPOSED

FINDING

Recovery still needed a fresh route from the cloud



RISK

Network latency could delay the switch or leave a stale route



DECISION

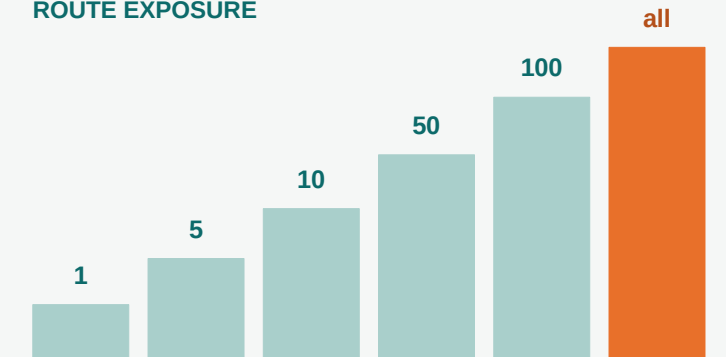
Delay release by one week, add two route-refresh points



CORRECTION

Request on signal recovery, refresh again at handover

ROUTE EXPOSURE



Each step held until the gate metrics stayed clean. Full launch followed the 100-route step.

The recovery gate was not weakened to keep the schedule. Release moved instead.

What I can attribute, and what I cannot

DIRECTLY OBSERVED

16.8%

shorter manual intervention
on flights where navigation
recovered

before



after



100+

controlled test flights before any
route was exposed

1 → 100

routes in staged rollout, then fleet-
wide launch

Measured on takeover logs over the rollout period,
on flights where navigation recovered.

What I would change

Build automatic bad-case capture first. Manual incident collection and no log return channel slowed every later validation.

DIRECTIONAL BUSINESS OUTCOME

+21%

active routes six months after launch

Why I do not claim this

Other technical and operational changes launched in the same period. Service metrics stayed within guardrails, but route growth has more than one cause.

NOT CLAIMED

Offline replay figures stayed out of this deck: they show potential, not observed online change.

WHERE THE EVIDENCE SITS

Operator capacity



Takeover duration
reversible recovery shipped

measured



Takeover count
envelope widened, alerts added

partial



Monitoring load
alert coverage extended

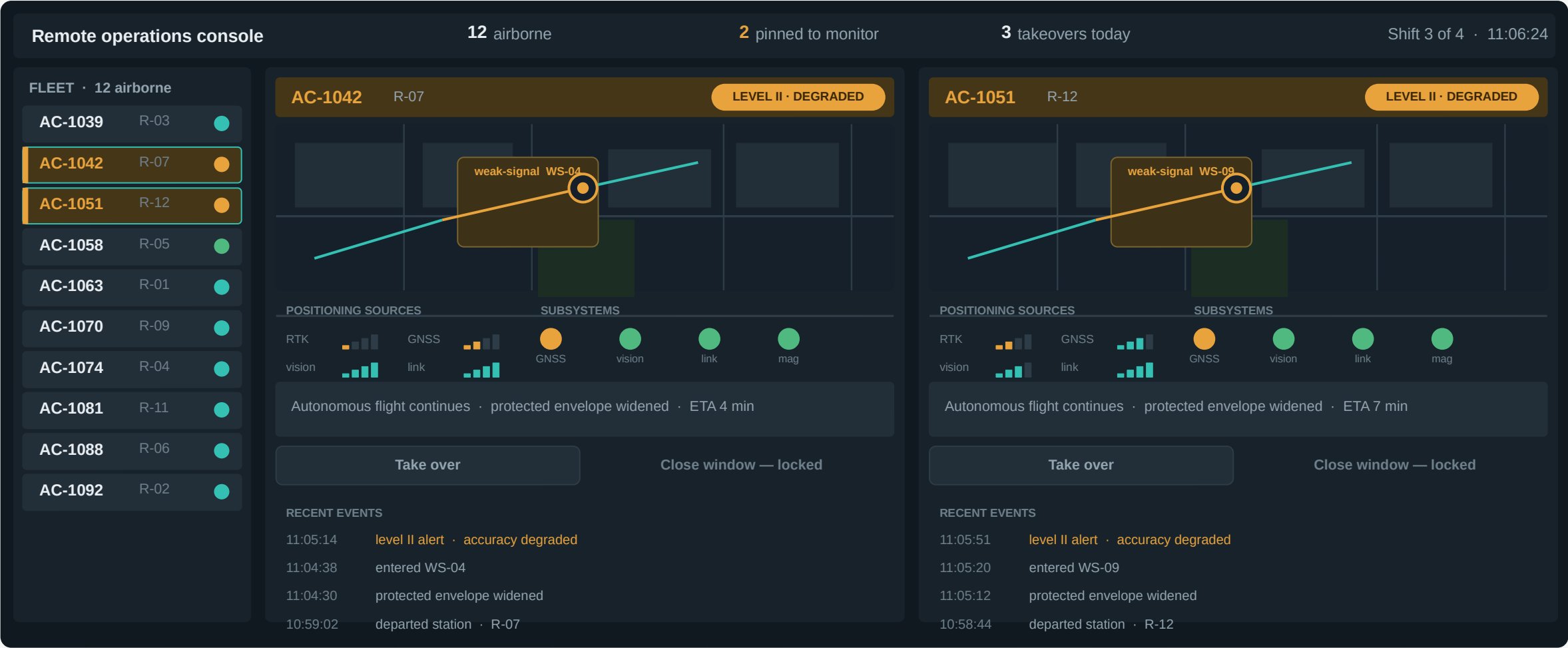
partial

One branch of the diagnosis is backed by a measured figure. The other two shipped, but I did not hold a clean number for them.

Product principle

In safety-critical systems with rare failures, observability and replay belong in scope from the start.

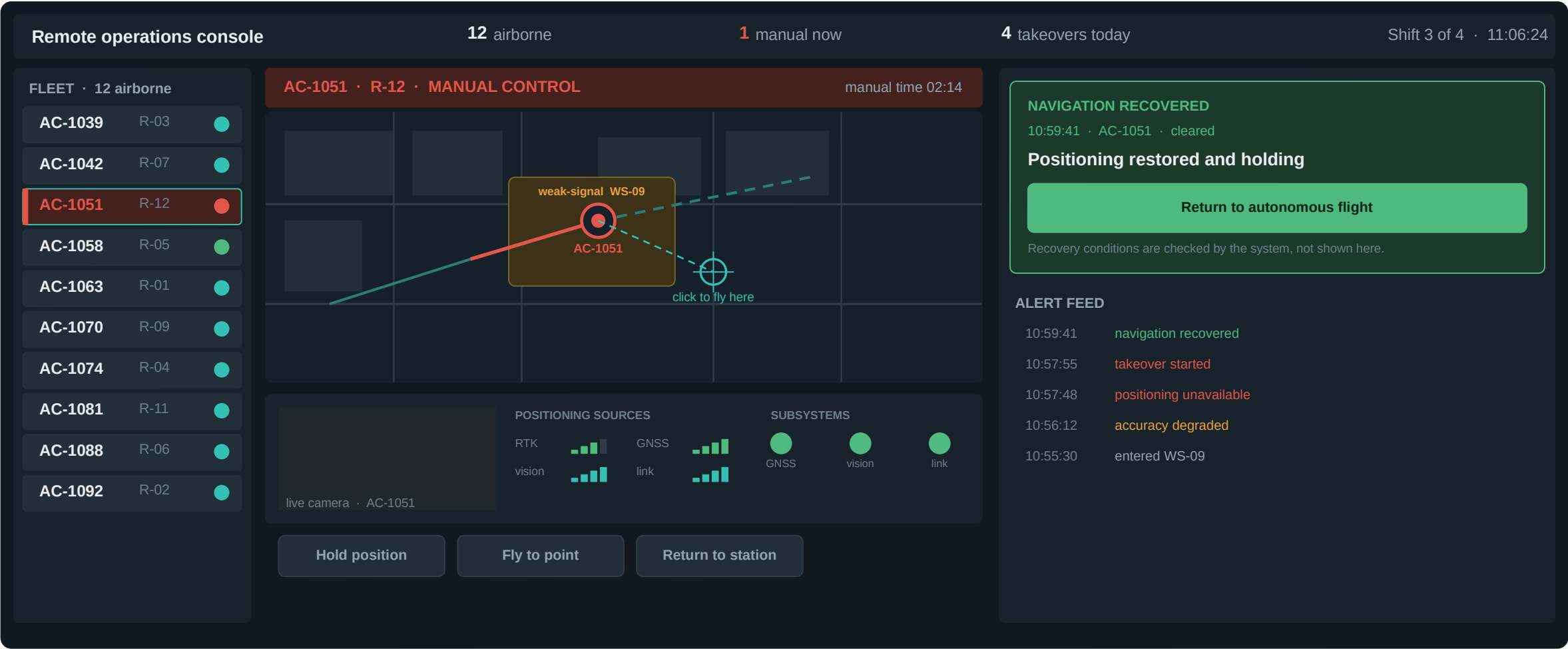
Forced monitor: two aircraft pinned, both still autonomous



Level II pins the window open until the alert clears. The operator must watch, but neither aircraft leaves autonomous flight — that is where operator capacity is saved.

Redrawn wireframe. IDs, routes, zones, codes and times are illustrative; no internal screens, fields or log samples are shown.

Manual takeover, and the one click that hands control back



Takeover no longer runs to landing. When navigation recovers the label turns green and one click returns the aircraft to autonomous flight.

Redrawn wireframe. IDs, routes, zones, codes and times are illustrative; no internal screens, fields or log samples are shown.